

The British Access Network

Bank of England crisis access from bill brokers to authorised money-market counterparties, 1847 to 1996

Ananya Besufekad
Digital History

June 2026

Abstract

The Bank of England's access network survived by changing form. Most individual firms changed, but the structure did not. Until the 1997 reform, the Bank still managed money-market stress through recognised intermediaries rather than through an open public market. In the Victorian crises this looked like a closed information and access club. A recurring set of firms appeared early, and the Bank's records showed strain before the public rupture. After 1914 the evidence is more cautious. It shows a closed access network more clearly than a closed information club, because later records tell us who the Bank dealt with more reliably than they tell us who saw trouble first. The continuity score used in this paper stays high from 1847 to 1996, then falls sharply at the 1997 reform, when the Bank widened its dealing list from recognised discount houses to a larger defined set of eligible institutions. That was a broader closed list, not access for the general public. We measure dated visibility, positioning, and institutional access. We do not claim that anyone knew a crisis in advance or profited from one.

Glossary

The paper uses a few financial terms, so I define them here before the argument begins. **Bill broker**: an early-nineteenth-century firm that matched lenders and borrowers of short IOUs (bills of exchange). **Discount house**: the bill broker's descendant, which buys ("discounts") short bills and holds them, funded by money lent overnight by the banks. **Acceptance house** (merchant bank): guarantees ("accepts") a bill for a distant borrower, lending its name not its cash. **Clearing bank**: a large ordinary deposit bank that lends spare cash to the discount houses. **Bank Rate**: the rate at which the Bank lent; raising it was its standard defence against a gold drain. **Open-market operation**: the central bank buying or selling securities to add or drain cash. **Lifeboat**: the nickname for the 1973 to 1975 rescue run by the Bank and the clearing banks.

1 Introduction

To the public, a financial crisis often appears as a headline. A famous firm fails, confidence breaks, and the panic seems to arrive all at once. For the people who traded money in London, that headline came later in the story. Before a crisis became public, bills were being refused, rates were rising, firms were looking for cash, and some institutions were already moving toward the Bank of England.

This paper asks who could move first. Was the City of London a closed access club, meaning a small recognised set of firms through which the Bank managed money-market stress? Or was it closer to an open market where any troubled institution could reach the central bank

on equal terms? If the City was a club in the Victorian period, did that system disappear as communication improved and modern central banking developed, or did it survive in a new form?

Research question. Across Britain’s crises from 1847 to 1996, did the Bank of England manage financial stress through a narrow recognised intermediary layer rather than an open public market? Did a stable inner circle sit at the front of the crisis information chain in the Victorian period? Did the 1866 telegraph shrink that advantage? And did the structure survive the First World War, from Victorian bill brokers and discount houses to interwar discount houses, acceptance houses, and later authorised money-market counterparties?

The answer. The Victorian City did behave like a closed information and access club. The same kinds of intermediaries recur at the front of the crisis chain, and the Bank’s defensive signal appears before the public rupture in every Victorian case. The 1866 transatlantic cable made prices travel faster, but it did not open access to the Bank. In the comparable transatlantic panics, the measured lead from the Bank’s signal to the public rupture grew rather than shrank. The price evidence does not show that early access produced clear profits, so the paper argues for an access advantage, not a proven profit advantage. After 1914, the structure survived in a more formal shape. The Bank kept working through recognised intermediaries until 1997, when it moved from the old discount-house list to a wider defined set of eligible counterparties.

The paper has two parts. Part I reconstructs the Victorian club and tests it against the telegraph and against share prices. Part II follows the access structure to 1996 and scores how much of it remained. This is a digital-history argument because the pattern only appears when separate digitised sources are placed on the same timeline. A contemporary observer might have seen one ledger, one newspaper, or one crisis. This project combines Bank transaction ledgers, a daily two-century interest-rate series, digitised newspapers, a specialist financial periodical, a 1906 bill dataset, interwar archive-based research, the Bank’s own institutional history, and a late-twentieth-century balance-sheet series. The digital work makes it possible to follow actors across sources that were never designed to be read together.

2 The actors and the access network

The argument follows a small cast of actors. The Bank of England sits at the centre. Around it are the firms that could reach the Bank, the banks that funded them, the newspapers that made stress public, and the wider public that usually learned later. The important point is that the individual names change more than the roles do. Table 1 sets out the cast, what each actor does, which source lets us see it, and why it matters to the argument.

The thread running down the middle of the table is the discount house. It begins as the bill broker the Bank gave a facility to in 1830, it is the spine of the network in every later case, and it ends as the authorised counterparty of 1996. The bill broker of 1830 and the authorised counterparty of 1996 occupy the same position in the network, right next to the Bank, even though almost none of the individual firms are the same.

3 What other scholars say

The paper speaks to two different ways of thinking about financial crises.

The closed-club side. Hopkins (1988), the open-access precursor to Cain and Hopkins’s *British Imperialism*, describes “gentlemanly capitalism” as a London elite whose advantage was relational. City “bankers and merchants ...were able to use their geographical location and social ties to make valuable political connections in the capital” (p. 6). Amini and Toms (2021) give

Table 1: The actors in the access network

Actor	Role in the network	Evidence that reveals them	Matters most	Why it matters
Bank of England	sets Bank Rate, lends at last resort, keeps the ledgers	ledgers, daily Bank Rate, Quarterly Bulletin	every period	the hub: whoever it deals with defines the network
bill brokers	match lenders and borrowers of short bills	1906 bill data, Bank 1967 history	early 19th c.	the original intermediaries, ancestors of the discount houses
discount houses	buy and hold short bills, funded by overnight call money	Victorian ledgers, 1906 data, 1931 archive, 1976-96 database	1830s to 1997	the spine of the network across the whole period
acceptance houses / merchant banks	guarantee (“accept”) bills for distant borrowers	Victorian context, 1906 data, 1931 archive	19th c. to 1930s	they screen borrowers and form part of the recognised layer
clearing banks	large deposit banks; supply call money; partner in rescues	1931 archive, 1978 lifeboat account	1900s onward	the recognised core that funds and helps organise support
secondary banks	1970s deposit-taking outsiders	Bank 1978 account	1970s	the rescued outsiders, who define the network by exclusion
specialist City press	report money-market strain for professionals	<i>The Bankers’ Magazine</i>	Victorian to interwar	shows stress was legible to insiders before the public
public, Parliament, newspapers	the broad public record	HMD newspapers, Hansard, failures	every period	the public clock, the comparison point for “before the public”
telegraph and communication	move price and news information	Hoag, Richmond Fed	1866 onward	could have opened access; it democratised prices, not access
authorised money-market counterparties	the Bank’s recognised dealing list	Bank database, statistics notes	1980s to 1996	the Victorian club, now a formal list

network evidence that interlocking directorates were private information conduits, giving some directors “a better informational advantage” (p. 497). Gorton and Ordoñez (2014) supply the theory that a crisis is an information event. Flandreau and Ugolini (2011) show Victorian crisis liquidity ran through the discount-house and bill-broker layer with privileged Bank access.

The efficient-market side. Hoag (2006) and the Richmond Fed (Fessenden 2018) show the 1866 cable cut the London to New York securities information lag “from ten days to zero” and compressed transatlantic price gaps from 5 to 10 per cent down to 2 to 3 per cent. **Two cautions.** Schneider (2022) shows the Bank’s 1866 lending was politically negotiated and deliberately ambiguous, not a transparent public rule. The Rothschild Archive’s own account of “Nathan and Waterloo” shows the famous insider-profit legend is “more legend than fact ...no question of trickery,” rooted in antisemitic propaganda, so we measure dated visibility and never assume profit.

The paper separates two things that are easy to confuse. One is price information. Did prices in London and New York line up more quickly after the telegraph? Hoag and the Richmond Fed show that they did. The other is institutional access. Did the Bank of England stop dealing through a recognised circle of firms? That is a different question. The evidence here suggests that price information became faster and more open, while access to the Bank remained mediated through recognised houses. In other words, the market could become more efficient in prices without becoming open in institutional access.

4 Why crises are the measurement points

A fair objection is that crises are unusual. Why study a financial system through its worst days? My answer is that crises make a hidden structure visible. In normal times, the access network is partly quiet. A discount house rolls over its bills. A clearing bank lends spare cash overnight. A trusted firm gets credit because everyone knows its name. These relationships matter, but they do not always leave a clear public trace.

A crisis makes them loud. When money tightens, the Bank’s records, its rate changes, its rescue operations, the newspapers, and the official documents all become denser and more datable at the same moment. Suddenly we can see who can reach the Bank, who moves early, who receives support, and whether that support is mediated through a recognised set of firms or open to anyone. The crisis acts as a stress test that lights up the wiring of the system.

For that reason, the paper treats crises as measurement points. The claim is not that the access network only existed during panics. The claim is that panics force private relationships into the record. By comparing several crises, the paper can ask whether the same structure keeps appearing.

5 Why these cases were chosen

The cases were not chosen at random. They vary along the dimensions that matter for the question, including period, crisis type, surviving evidence, communication technology, and how clearly we can observe who reached the Bank.

Part I, the Victorian crises. The paper uses 1847, 1857, 1866, 1890, 1907, and 1914 because they are the major British crises of the press and telegraph era, and because together they let the paper compare crises that grew at home (endogenous) with shocks that arrived from outside (external or geopolitical). The mix is deliberate. 1847, 1857, 1866, and 1914 have Bank of England transaction-ledger evidence, which is firm-level and lets us name the intermediaries at the window. 1890, the Baring crisis, has strong daily account-book and rescue evidence even

though no transaction ledger survives, so it enters the lead-time comparison but not the firm-level recurrence. 1907 is included mainly as a post-cable transatlantic comparison to 1857, even though its evidence is thinner and rests largely on Bank Rate. 1873 is used where Bank Rate and public-crisis evidence allow, and it is flagged as a weaker case because its source coverage is thin.

Part II, the post-1914 cases. Here the cases are chosen to test whether the same access structure survived. 1906 is included as a prewar structural baseline, a snapshot of the bill market just before 1914. 1931 is the closest major post-1914 British financial crisis for which a recent archival reconstruction of the Bank's operations exists. 1973 to 1975, the secondary banking crisis, is the major later British rescue case before the modern era of central banking, and it comes with named institutions and an official rescue structure. The 1976 to 1996 discount-house data show whether the recognised intermediary layer was still part of ordinary Bank operations in the decades before reform. And 1997 is included as the institutional break, the year the Bank stopped dealing exclusively with members of the London Discount Market Association.

Together, these cases cover about 150 years. They include crises that began inside London finance and crises that arrived from abroad. They also sit on both sides of the 1866 transatlantic cable. The evidence changes from firm-level ledgers to aggregate series and official histories, but that is part of the point. If a similar access structure appears across such different kinds of sources, the continuity claim is stronger than it would be from one crisis or one dataset.

6 Sources and source criticism

Every claim in the paper is tied to a source, and the claim is kept within what the source can actually show. Table 2 explains where each dataset came from, what it contains, why it is used, how it is analysed. The detailed criticism follows the table in paragraph form.

Table 2: Sources, uses, and methods

Source / dataset	Where obtained	What it contains	Why this source is needed	Analysis method
A. BoE lender-of-last-resort transaction ledgers	project data, Anson et al. 2017 (BoE historical dataset)	counterparty names, transaction dates, lending amounts, for crises with ledgers	firm-level Victorian access and the recurring inner circle	counterparty classification, recurrence analysis, early-access timing, access scoring
B. Daily Bank Rate (Millennium dataset)	BoE <i>A Millennium of Macroeconomic Data</i>	the daily Bank Rate, 1833 onward	a common market-facing stress signal for every crisis	lead-time calculation against the public rupture
C. HMD newspapers, Hansard, <i>Bankers' Magazine</i>	British Library HMD, Hansard, Internet Archive	dated public and specialist textual traces	the public and specialist visibility clocks	text mining, keyword search, public-clock validation, hand checking
D. Yale Investor's Monthly Manual	Yale ICF, project processed data	monthly listed financial-sector prices	the benefit and profit proxy test	check whether listed-financial prices fall before public ruptures
E. 1906 sterling bills	Accominotti, Lucena-Piquero, Ugolini (arXiv)	reported counts from 23,493 rediscounted bills	the prewar bill-market access funnel	concentration and funnel analysis from published counts
F. Römer 1931	Römer 2025, <i>Economic History Review</i>	archival reconstruction of 1931 Bank operations, OMO, discount and acceptance houses	the interwar crisis access network	source-derived actor-flow network, semi-quantitative amounts
G. BoE 1967 discount-market history	BoE <i>Quarterly Bulletin</i> 1967	institutional history of bill brokers, discount houses, the LDMA, Bank facilities	long-run institutional continuity	qualitative institutional tracing
H. BoE 1978 secondary banking crisis	BoE <i>Quarterly Bulletin</i> 1978 Q2	official account of the lifeboat, support amounts, Control Committee, named institutions	the later British selective rescue network	source-derived rescue network
I. BoE discount-house database, 1976 to 1996	BoE statistical database	monthly aggregate balance sheets, incl. Bank deposits placed with discount houses	evidence that the Bank still used the houses before 1997	time-series summary: non-zero months, mean, median, max, trend, volatility, taper

A. Bank of England lender-of-last-resort transaction ledgers. This is the strongest source for the Victorian part of the paper because it is closest to the action being studied. The data come from the Bank of England historical dataset discussed by Anson et al. (2017) and used in the project pipeline as transaction-level crisis lending records. The records were curated from Bank archival ledgers, so they preserve the names of counterparties, dates of transactions, and amounts lent in the crises for which ledgers are available. Their reliability comes from their institutional origin. They were working records of the Bank, not later memoirs or newspaper summaries. They are therefore strong evidence for who appeared at the Bank’s window and when. Their weakness is coverage. They do not cover every crisis in this paper, and they do not explain motives. A firm appearing early in the ledger shows access and timing, not private knowledge, intention, or profit. The counterparty classifications also depend on later cleaning and classification work, so the paper uses them to identify broad actor types rather than to claim perfect firm biographies.

B. Daily Bank Rate from *A Millennium of Macroeconomic Data*. The daily Bank Rate series comes from the Bank of England’s official long-run macroeconomic database. It is reliable for this paper because it is an official, continuous, standardised series produced by the institution whose policy signal is being measured. The series is not a private ledger. It does not tell us which firm borrowed or which banker had information. Its value is different. It gives a common daily clock across crises, so the paper can compare when the Bank began to signal stress through rate policy. The weakness is interpretive. A rise in Bank Rate can reflect several pressures, including gold outflows, exchange-rate defence, and general market stress. For that reason, the paper treats Bank Rate as a market-facing stress signal, not as proof that the Bank had identified a specific failing firm.

C. HMD newspapers, Hansard, and *The Bankers’ Magazine*. These textual sources were used to build the public and specialist visibility clocks. The public newspaper material comes from the British Library’s digitised newspaper corpus through the HMD material used in the project. Hansard gives the parliamentary record, and *The Bankers’ Magazine*, accessed through digitised copies on the Internet Archive, provides specialist City commentary. These sources are reliable for dating visibility because they are contemporary printed records. They show when a topic became speakable in public, Parliament, or the specialist press. They are weaker as evidence for the whole public sphere. Digitised newspapers are shaped by survival, scanning, and OCR quality. Keyword search can miss articles with different wording, and *The Bankers’ Magazine* is monthly, so it cannot date daily changes as precisely as the Bank Rate. For that reason, the paper uses text mining as a visibility marker and then checks important dates by reading the hits, rather than treating raw keyword counts as complete public opinion.

D. Yale *Investor’s Monthly Manual*. The price data come from the Yale International Center for Finance version of the *Investor’s Monthly Manual*, processed in the project into a monthly listed-financial-sector index. The source is useful because it is a contemporary market-price record, not a later interpretation. It lets the paper test whether listed financial shares moved before public crisis ruptures. That is why it is used as a benefit proxy. Its weakness is that it is monthly and sector-level. It cannot show daily trading, private positions, or firm-level profits of unlisted discount houses. Several of the most important access-network institutions were not ordinary listed firms. A negative result in this source therefore means only that no broad listed-sector price footprint is visible. It does not prove that no individual insider benefited.

E. The 1906 sterling bills reconstruction. The 1906 bill-market evidence comes from Accominotti, Lucena-Piquero, and Ugolini’s reconstruction of sterling bills, read through their arXiv version and supporting source notes. It is a scholarly reconstruction from bill-market records, and the paper uses the published counts of drawers, acceptors, and discounters rather

than claiming direct access to every underlying archival entry. The source is reliable for the structure of the prewar bill market because it is built from a large sample of 23,493 rediscounted bills and appears in peer-reviewed or near-publication form. Its main weakness for this paper is that 1906 is not a crisis year. It shows the normal funnel through which access was organised, not how firms behaved under panic conditions. It also gives aggregate concentration measures more clearly than it gives a complete downloadable firm-level dataset for independent reclassification.

F. Römer’s reconstruction of the 1931 crisis. The 1931 evidence comes from Römer’s 2025 *Economic History Review* article, which reconstructs the Bank’s operations from archival material. It is reliable because it is a recent peer-reviewed study based on Bank archival records, and it provides a rare bridge between the Victorian cases and the later twentieth century. This paper uses it for actor-flow reconstruction and semi-quantitative support amounts. The weakness is that the raw transaction data behind Römer’s reconstruction are not released as a dataset for this project. That means the paper cannot independently rebuild every 1931 transaction or produce a full firm-level network. The 1931 diagram is therefore labelled as source-derived. It represents the relationships described in the article rather than a newly extracted archival database.

G. Bank of England 1967 discount-market history. The 1967 *Quarterly Bulletin* article is an official Bank of England account of the London discount market. It is used because it explains the institutional lineage from bill brokers to discount houses, including the Bank’s 1830 facility and the later London Discount Market Association structure. Its reliability lies in its institutional knowledge. The Bank was describing a market in which it directly participated. Its weakness is also its institutional authorship. It is not an outside critique and may present the Bank’s arrangements as more orderly or natural than they looked to outsiders. The paper therefore uses it for institutional facts and lineage, not for hidden politics or for judging whether the system was fair.

H. Bank of England 1978 account of the secondary banking crisis. The 1978 *Quarterly Bulletin* article is the Bank’s official account of the lifeboat rescue. It contains the Control Committee structure, named institutions, support amounts, and the criteria for inclusion. It is reliable for reconstructing the formal rescue architecture because it comes from the organiser of the operation and gives details that are hard to obtain elsewhere. Its weakness is that it is a rescuer’s account of its own rescue. It may justify decisions, understate conflict, or omit private negotiations. For that reason, the paper uses it to show selective access, vetted entry, and the Bank-clearing-bank network, but not to claim a complete political history of the 1970s crisis.

I. Bank of England discount-house database, 1976 to 1996. The late-twentieth-century discount-house data come from the Bank of England statistical database. They are official structured balance-sheet data and are reliable for showing whether the Bank still placed sterling deposits with the discount-house sector before the 1997 reform. The series was processed into monthly summary statistics, including non-zero months, mean, median, maximum, volatility, and the pre-reform taper. Its main weakness is aggregation. It is a sector-level series, not a firm-by-firm ledger, and under the available current series codes the exposed coverage begins in 1986 rather than 1976. It can show that the discount-house layer remained in active use, but it cannot show which authorised counterparty received each deposit or why a particular placement was made.

Across these sources, three limits remain. Most post-1914 evidence is institutional or aggregate, so Part II argues continuity of the network, not that identical firms ran the whole period. The named threads, especially Union Discount and Seccombe Marshall & Campion, are illustrative rather than a complete roster. No source here can prove private knowledge or private profit.

7 Method

The paper uses several methods because the sources are different kinds of evidence. A transaction ledger, a newspaper, a monthly price index, and an official history cannot be read in the same way. Each method below is matched to the source it can handle.

- **Transaction-ledger analysis** is used on the Victorian Bank ledgers (source A). It classifies counterparties into types, measures how often each firm recurs across crises, and times when each first appears at the Bank’s window, which produces the recurring-inner-circle evidence in Part I.
- **Network analysis** turns that recurrence evidence into a graph. The named institutions are linked whenever they sit at the Bank in the same period, and standard graph measures (eigenvector and betweenness centrality) then identify which firm is most central to the access network across the whole period. This makes the word network in the title a measured object rather than a metaphor.
- **Lead-time analysis** is used on the daily Bank Rate (source B). For each crisis it dates the Bank’s first sustained defensive rate rise and measures the gap to the public rupture.
- **Text mining** is used on the newspapers, Hansard, and the specialist periodical (source C). Keyword searches over dated articles locate when a crisis became visible in the public and specialist record, which is the public clock the lead time is measured against.
- **Concentration and funnel analysis** is used on the 1906 bill counts (source E), comparing the numbers of drawers, acceptors, and discounters and the concentration of each layer.
- **Source-derived actor-flow networks** are used for 1931 and 1973 (sources F and H). Because the underlying records are not released, these are structured diagrams that encode the relationships the articles describe, and they are labelled as such.
- **Time-series summary** is used on the 1976 to 1996 discount-house data (source I), reporting how often and how much the Bank placed with the houses, plus the trend and any taper before reform.
- **Continuity-index scoring** ties the periods together, scoring six yes-or-no criteria per period (named institutions visible; formal eligibility or recognised membership; Bank access restricted or mediated; crisis support routed through intermediaries; quantitative evidence available; direct crisis evidence), which sum to a score from 0 to 6 with a confidence weight. The criteria separate an open market from a mediated one: an open market would not need named institutions, a formal eligibility list, restricted access, or a fixed intermediary route, so it would score low on the first four. The last two are about how far we can trust the reading, not the structure, which is why each period carries a confidence weight. The index is not a truth machine; it is a transparent scoring device a reader can disagree with criterion by criterion.
- **A price-proxy test** is used on the Yale IMM index (source D), asking whether listed-financial prices fell before each public rupture, as a check on whether early access showed up as anything a market could see.

Throughout we say “visibility,” “positioning,” and “access,” not “knew” or “profited.”

Part I. The Victorian information and access club, 1847 to 1914

8 The information chain and the insider-to-public lead

The first test lines up three clocks for each crisis. The first clock is the Bank or market signal, such as a sustained rise in Bank Rate. The second is the specialist financial record. The third is the public rupture, the moment when the crisis became visible as a public event. Table 3 shows a consistent pattern. A market-facing signal comes before the public rupture in every case. The lead is longest when trouble grows inside London finance, about 277 days in 1847 and 142 days in 1890. It is shorter when the shock comes from abroad, and almost disappears in 1914, when war froze the system overnight. The Bank's records did not prove that anyone knew the future. They did show that stress became measurable before it became public.

Table 3: The seven crises and the insider-to-public lead

Crisis	Origin	Public rupture	Earliest market-facing signal	Lead	What it supports
1847	endogenous	18 Oct 1847	14 Jan 1847 (Bank Rate climb begins)	277 d	a long, visible build-up at home
1857	external (US)	9 Nov 1857	15 Oct 1857 (Bank Rate); window from 6 Sep 1865 (Bank Rate high); ledger 4 Mar 1866	25 d	shorter lead for an imported shock
1866	endogenous	11 May 1866	25 Sep 1873 (Bank Rate)	288 d (run-up 8)	a long build-up, but a sudden public break
1873	external	7 Nov 1873	26 Jun 1890 (Bank Rate)	43 d	imported shock, short lead
1890	endogenous	15 Nov 1890	15 Aug 1907 (Bank Rate)	142 d	home-grown, longer lead
1907	external (US)	22 Oct 1907	30 Jul 1914 (Bank Rate)	68 d	imported shock
1914	external (war)	31 Jul 1914		1 d	a war froze the system overnight

The lead column is the heart of the Victorian claim. The market-facing record moves before the public record in every crisis, and it moves much earlier when the trouble is home-grown. This is what early positioning looks like in dated evidence. It is enough to show a gap between market visibility and public visibility, without claiming that anyone could predict the crisis or make easy profits from it.

9 The telegraph test: did the cable open the club?

The telegraph gives the paper a useful test. If the City's advantage was only about faster news, the 1866 cable should have weakened it. The literature shows that the cable did collapse the price lag between London and New York. But the Bank's lead over the public rupture did not shrink. The cleanest comparison is between two United States-origin panics, one before the cable and one after it. In 1857 the Bank's defensive signal came 25 days before the public rupture. In 1907, four decades after the cable, it came 68 days before the rupture (Figure 1). The lead grew rather than shrank. This suggests that the telegraph and the club worked through different channels. The telegraph opened price information. It did not decide who could stand at the Bank's discount window.

The telegraph test: did the 1866 cable shrink the insider-to-public lead? For the transatlantic crises, no.

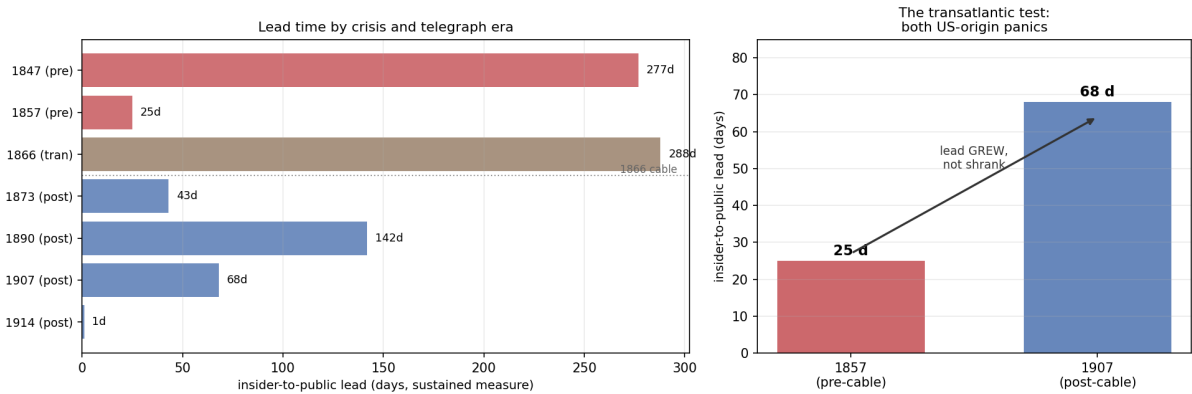


Figure 1: The insider-to-public lead by crisis and telegraph era, and the transatlantic pair. The cable cut the price lag, but the lead did not shrink: 1857 (25 days) to 1907 (68 days).

10 The recurring inner circle

The transaction ledgers show that the front of the crisis chain was not random (Figure 2). Overend, Gurney appears in 1847, 1857, and 1866, until it fails in 1866. R. Raphael & Sons and Frith Sands also appear across the mid-century crises. Of the ten intermediaries with the highest access scores, all ten reached the Bank's window early. These recurring houses were mainly discount houses, bill brokers, and merchant banks. Those are exactly the kinds of firms named in the historical literature on the City. This does not look like an open public market. It looks like a narrow recognised layer.

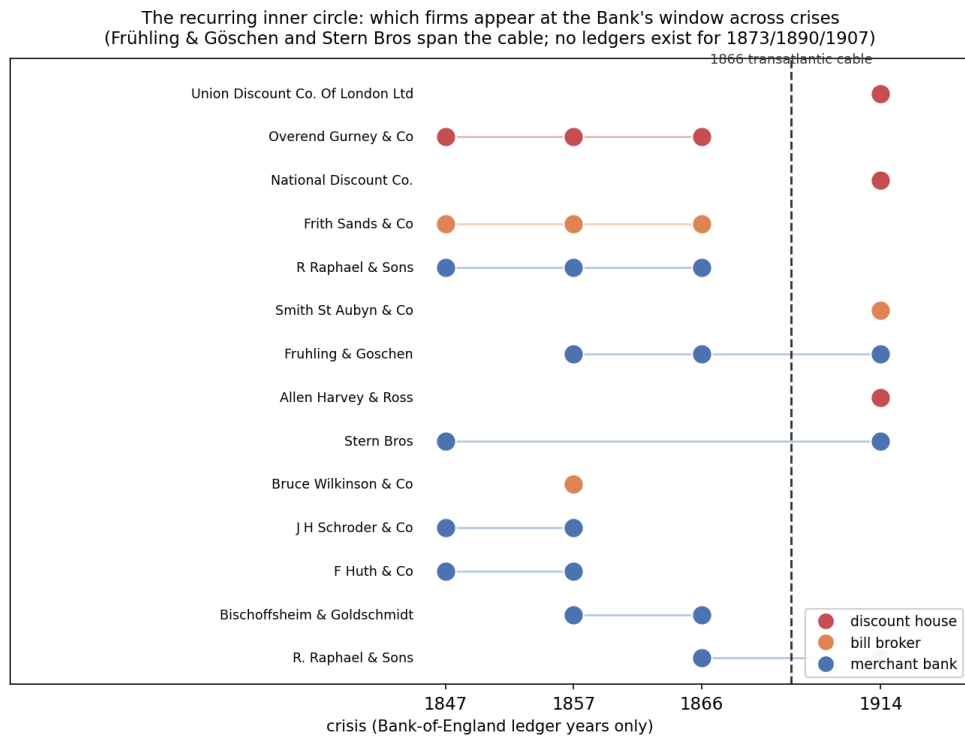


Figure 2: Which firms appear at the Bank's window across the four Victorian ledger crises. A small set recurs, and Frühling & Göschen and Stern Bros reappear in 1914.

The access network drawn as a graph

Section 10 reads recurrence off a list. The same evidence can be turned into a network and measured directly, which is a fairer test of the word network in this paper’s title. Each named institution documented above is joined to every crisis period in which it sits at the Bank, which gives a bipartite institution-to-period graph spanning 1847 to 1996. Projecting that graph onto the institutions themselves, so that two firms are linked whenever they sit at the Bank in the same period, gives the co-appearance network in Figure 2b. Links are weighted by the number of shared periods. The graph has eighteen named institutions and fifty-three links, and exactly half of the eighteen firms are discount houses.

Eigenvector centrality, the standard measure of how embedded a node is in the well-connected core of a graph, places a single firm at the centre of the whole structure (Table 3b). Union Discount Co. scores 0.49, clear of every other firm, and it also has by far the highest betweenness, 0.52. It is the one node that bridges the Victorian acceptance cluster of the 1860s to the authorised money-market counterparties of 1996, the same firm that was the largest discounter in 1906 and that appears in the Victorian ledgers and on the 1996 list. This is the quantitative version of the paper’s central claim. The most structurally central institution across a century and a half of crises is a discount house, and the discount houses form the connected spine of the graph while the rescued secondary banks of the 1970s sit on its outer edge. Because the network is built from documented named firms rather than raw transaction matching, it measures the structure the sources record rather than proving a hidden one, but the measure and the narrative point the same way.

Table 3b: The most central institutions in the access network

Institution	Actor type	Periods	Eigenvector	Betweenness
Union Discount Co.	discount house	5	0.49	0.52
National Discount Co.	discount house	2	0.36	0.01
Barings	acceptance house	3	0.36	0.01
Rothschilds	acceptance house	2	0.36	0.00
Overend, Gurney & Co.	discount house	3	0.23	0.00

11 The benefit test: did early access show up in prices?

Early access is not the same thing as profit. This matters because stories about financial insiders often slide too quickly from “they heard early” to “they profited.” The Rothschild Archive’s correction of the Waterloo legend is a warning against that move. To check whether early access showed up in prices, the paper uses the Yale IMM monthly index of listed financial institutions. If listed financial firms were being repriced before the public rupture, the index should fall. It does not. In 1873, 1890, and 1907, the sector was still rising into the public rupture, by about 5, 21, and 30 per cent over the prior six months (Figure 3). The listed market was not pricing in these crises in advance. This is why the paper argues for a club of access and visibility, not a proven profit advantage. The source also has a limit. The IMM index is monthly and sector-level, not firm-level, and several key discount houses were not listed. A proper profit test would require sources this project does not have.

The access network as a graph: institutions linked when they sit at the Bank in a shared period
Node size = eigenvector centrality. Most central: Union Discount Co. (5 periods). Edge width = shared periods.

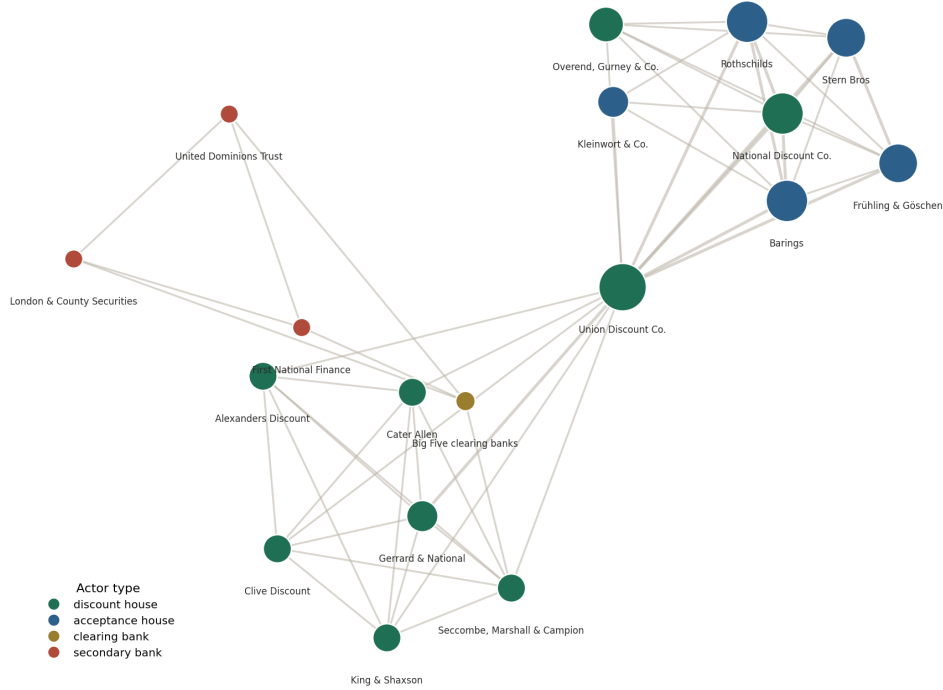


Figure 2b: The recurring institutions as a co-appearance graph: a node for each named firm, an edge when two firms sit at the Bank in the same period, node size scaled to eigenvector centrality and edge width to shared periods. Union Discount Co. is the central, bridging node. Built with NetworkX from the named-recurrence evidence; it describes structural position across periods, not transaction volume.

Market-price proxy (Yale IMM, listed financials): the sector was still RISING into each public rupture
-- no sign of early repricing, so we make no claim of insider profit

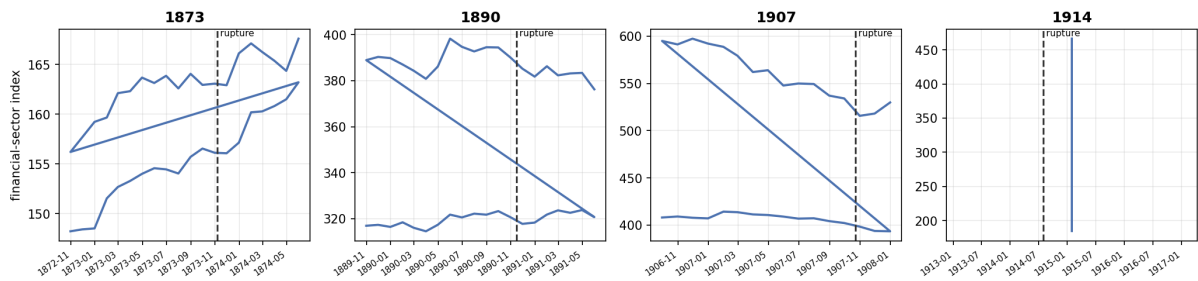


Figure 3: The listed-financial-sector index around each post-1869 rupture. The sector rose into every rupture, so there is no market-price sign of early repricing, and we make no profit claim.

Part II. Did the structure survive? The access network, 1906 to 1996

12 From the Victorian club to the access network

The second part of the paper asks whether the structure survived after the Victorian period. It would be too easy to jump from the ledgers to a modern rescue and call the pattern permanent. That would flatten history. Law, technology, communication, and the banking system all changed. The question is more careful. Did the Bank keep working through a recognised intermediary layer? The later evidence usually cannot show who saw trouble first, so Part II stops using the stronger phrase “information club” and uses the more cautious phrase “access network.”

13 The 1906 bill market

The 1906 bill market shows the access network in a calm year. A borrower anywhere in the world, called the drawer, could write a short-term bill. To make that bill trusted in London, an acceptance house guaranteed it. A discount house then bought the bill and could take it toward the Bank. The important feature is the narrowing funnel (Figure 4). On roughly 23,493 bills rediscounted by the Bank in 1906, there were about 3,554 drawers and 1,439 acceptors, but only 145 discounters. The closer we get to the Bank, the more concentrated the market becomes. The top fifteen acceptors handled about 35 per cent of accepting, while the top fifteen discounters handled more than 70 per cent of discounting. The largest discounter was Union Discount, a name that also appears in the Victorian ledgers. This is not a crisis case, so it does not prove crisis behaviour. It shows the ordinary structure that crisis support would later travel through.

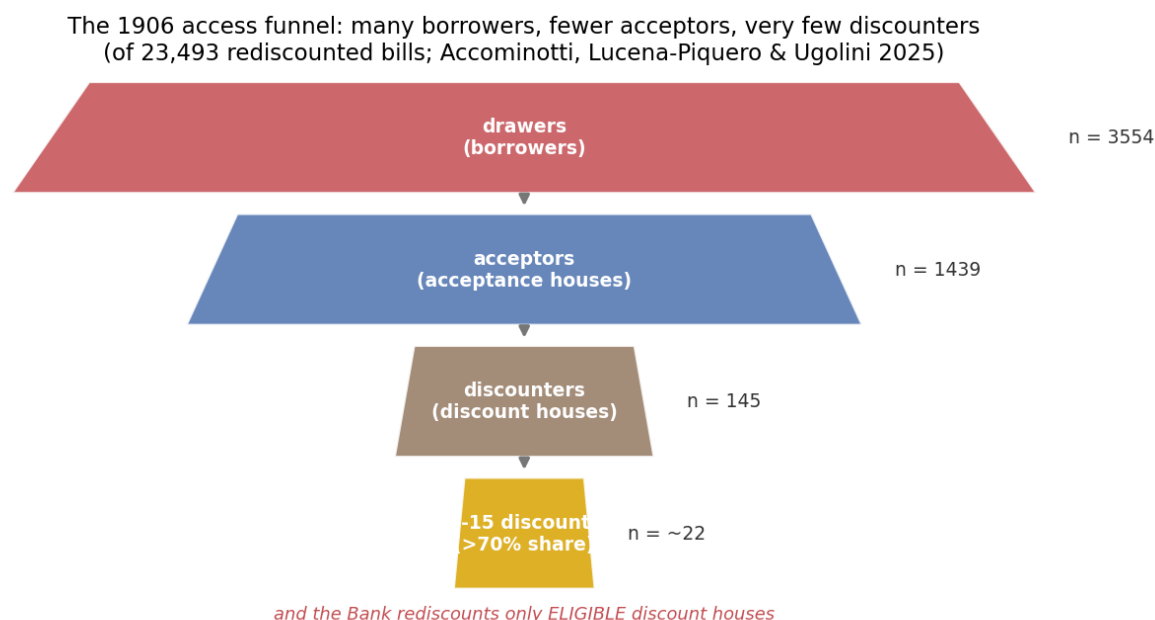
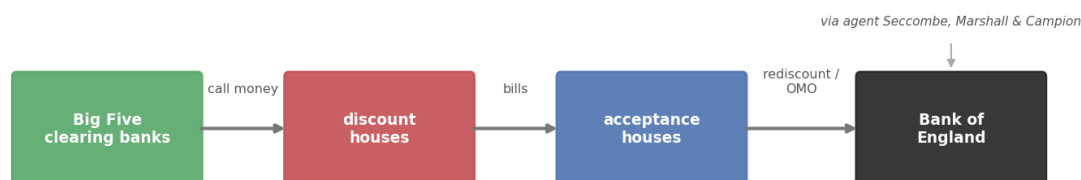


Figure 4: The 1906 access funnel: many borrowers, fewer acceptors, very few discounters, and only eligible houses at the Bank. Source values from Accominotti et al. (2025).

14 The 1931 crisis

The 1931 sterling crisis is the interwar test. Gold drained from London, the Bank raised Bank Rate, and Britain left the gold standard in September. Römer’s reconstruction from the Bank’s archive shows a familiar tiered system (Figure 5). The Big Five clearing banks, Barclays, Lloyds, Midland, Westminster, and National Provincial, lent call money to the discount houses. The discount houses held bills. The acceptance houses guaranteed bills. The Bank sat at the top. When the Bank added liquidity, it bought about 25 million pounds of Treasury bills through open-market operations routed through the discount houses. Its market bill transactions also ran through its special agent, Secombe, Marshall & Campion, a name that appears again in the 1996 authorised list. Römer writes that the 1931 response was similar to 1914 because the Bank purchased large amounts of bills from both acceptance houses and banks. The interwar Bank was still supplying liquidity through recognised market layers, not through open public access. Because Römer’s raw archive data are not released, this section uses his peer-reviewed reconstruction rather than a full transaction dataset.

1931: the Bank’s support flowed through the intermediary tier (source-derived from Römer 2025)



The Bank added liquidity by buying about £25m of Treasury bills from the discount houses (open-market operations). It did not lend to firms or the public directly.

Figure 5: The 1931 flow of support: clearing banks to discount houses to bills and acceptances to the Bank’s open-market operations. A source-derived diagram from Römer (2025).

15 The continuity of the discount market

The Bank’s own 1967 history is blunt about the lineage. “The origins of the discount market lie with the bill brokers,” it says, and “in 1830, the Bank decided to extend to certain brokers a facility for discounting bills.” By the twentieth century “the market effectively consisted of the eleven houses which had discount facilities at the Bank of England and thus had access to the Bank as lender of last resort.” The London Discount Market Association, the formal club of these houses, was founded in 1944, with a committee active since 1914. The Victorian inner circle was carried forward as a recognised group with a name, a facility, and privileged access to the Bank. Because this is the Bank’s own institutional history, it is reliable on the structure of the market but is not an outside critique of it.

16 The 1973 to 1975 secondary banking crisis

A group of secondary or fringe banks had grown by taking deposits and lending into a property boom, and when property fell they began to fail, starting with London and County Securities in late 1973. The rescue, nicknamed the lifeboat, was run by a Control Committee of the Bank of England and the English and Scottish clearing banks, chaired by the Bank’s Deputy Governor (Figure 6). The clearing banks carried most of the risk and the Bank 10 per cent, and each

rescued company was assigned a clearing-bank “related bank.” Support peaked at 1,285 million pounds in March 1975, and about 26 secondary banks were taken in (including the Stern Group, First National Finance Corporation, and United Dominions Trust), while a few large cases such as Slater Walker were handled by the Bank on its own account. The key question is whether this was broad public access or selective network rescue. The Bank’s account is clear that entry was vetted. A company qualified only if it “exhibited sufficient banking characteristics to justify inclusion in the scheme” and “had attracted a significant level of deposits from the public.” Most approved companies were identified within the first few months. This was not an open facility. It was the recognised core of British banking, the Bank and the clearing banks, rescuing a finite set of outsiders. The membership was new, but the logic was familiar. Because the source is the Bank’s own account, it is strongest on institutional structure and weakest as an independent critique of the Bank’s choices.

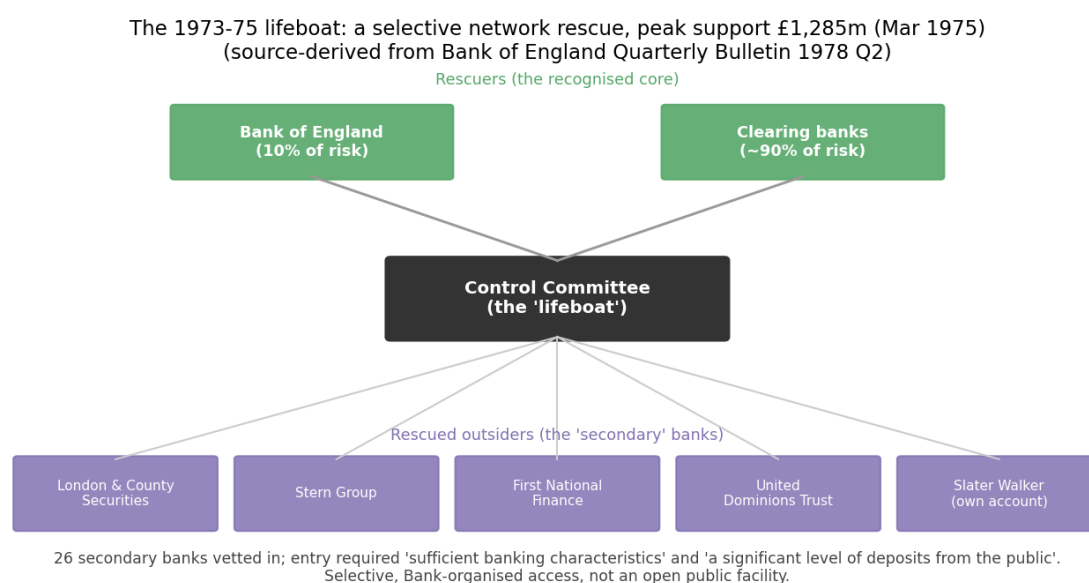


Figure 6: The lifeboat as a selective network rescue. The Bank and the clearing banks form the Control Committee that vets and supports a finite set of secondary banks. A source-derived diagram from the Bank’s 1978 account.

17 Discount houses, 1976 to 1996

The Bank’s data show the discount houses were no relic (Figure 7). One series records the sterling deposits the Bank itself placed with them. Over the 124 months from September 1986 to December 1996, that figure is non-zero in 105 months (a share of 85 per cent), with a mean of about 254 million pounds, a median of about 170 million, and a maximum of 1,930 million. The series is volatile (its coefficient of variation is 1.24) with occasional large spikes, which reads as routine but lumpy use of the houses rather than a single exceptional episode. There is a mild downward drift, and the last twelve months average about 217 million against about 258 million earlier, so usage tapers somewhat as the 1997 reform approaches, but the houses are still clearly in active use through the mid-1990s. Because the data are a sector aggregate, they show the continuing use of discount houses as a layer, not which firm received which deposit.

The structure was by then a formal list. The eleven LDMA houses of 1945 had consolidated into seven authorised money-market counterparties by 1996 (Figure 8): Alexanders, Cater Allen,

Clive Discount, Gerrard & National, King & Shaxson, Seccombe Marshall & Campion, and Union Discount. Union Discount was the largest discounter of 1906 and appears in the Victorian ledgers; Seccombe, Marshall & Campion was the Bank’s bill agent in 1931. The Bank’s notes say it operated “primarily through the discount houses” and “principally through members of the LDMA,” and that it “ceased to deal exclusively with members of the LDMA” only on 3 March 1997, when it widened its operations to a larger defined set of eligible counterparties rather than to the public.

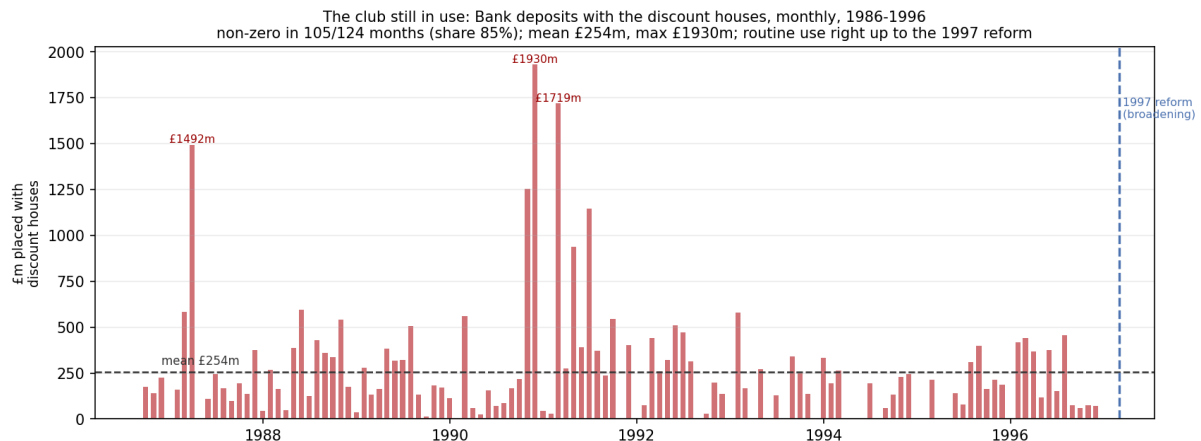


Figure 7: The Bank’s sterling deposits placed with the discount houses, monthly. Non-zero in 85 per cent of months, volatile, with a mild taper before the 1997 reform. Computed from the Bank’s database series RPMATJD.

1996: the Bank operated through seven authorised money-market counterparties (red = a name that also appears earlier in the network; the club, now a formal list)

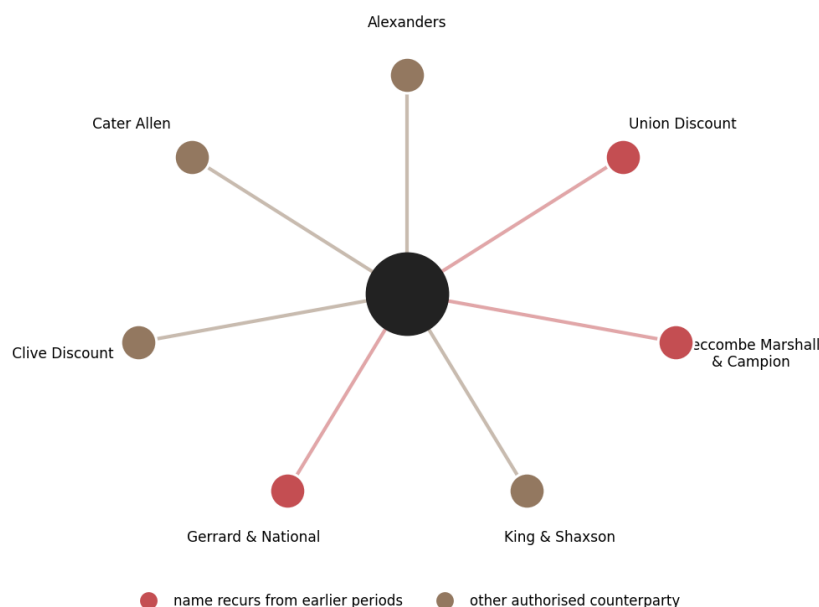


Figure 8: The seven authorised money-market counterparties the Bank operated through in 1996. Names in red also appear earlier in the network. The club is now a formal list.

18 Interpretation

The full arc is now a scored argument rather than a general impression (Figure 9, Table 4).

The access structure scores high from 1847 to 1996; exclusive discount-house dealing ends at the 1997 reform

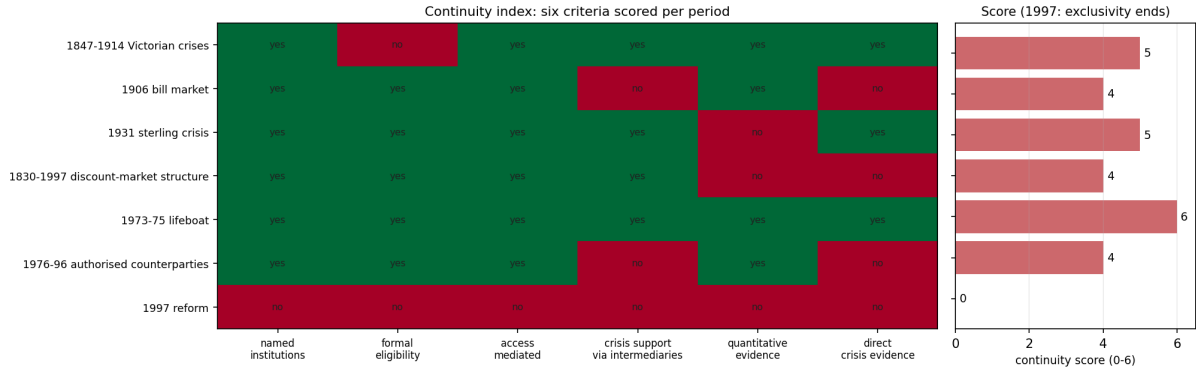


Figure 9: Six criteria scored per period, with the total continuity score. The structure scores 4 to 6 out of 6 from 1847 to 1996. The score for exclusive discount-house dealing falls to 0 at the 1997 reform, when access broadened from the recognised discount houses to a wider defined set of eligible counterparties (not to the general public).

Table 4: The continuity index

Period	Continuity score (0 to 6)	Confidence
1847 to 1914 Victorian crises	5	0.95
1906 bill market	4	0.90
1931 sterling crisis	5	0.80
1830 to 1997 discount-market structure	4	0.85
1973 to 1975 lifeboat	6	0.85
1976 to 1996 authorised counterparties	4	0.90
1997 reform	0	0.70

Table 5: The questions and the verdicts

Prediction	What we find	Verdict
The 1866 cable shrinks the insider-to-public lead	the transatlantic lead grew, 1857 (25 d) to 1907 (68 d)	against democratization
Early access shows up as profit in listed prices	listed financials rose into every rupture, no footprint	not supported, no profit claim
Support stays routed through recognised intermediaries after 1914	true in 1906, 1931, 1973 to 75, 1976 to 96 (index 4 to 6)	institutional continuity
The same firms run the whole period	only a few names recur, membership turns over	not supported (no sameness claim)
Access becomes broad and open to the public	the exclusive discount-house route ended only at the 1997 reform, and access then broadened to a wider defined set of eligible counterparties, not to the public	partial broadening in 1997

The named evidence separates two claims that should not be confused (Figure 10). A few firms do carry across the wars. Union Discount appears in five periods, and Seccombe, Marshall & Campion appears in two. But most names change. Overend Gurney fails in 1866. The secondary banks of the 1970s are outsiders being rescued, not members of the old club. The stronger continuity is therefore not any single firm. It is the actor type. A recognised discount-house and acceptance-house layer is present from the 1830s to 1997. The form changes, but the principle persists.



Figure 10: A few named firms carry across periods, but the strongest continuity is the actor type (gold), not any single firm. The named firms are illustrative, not proof that the same firms ran everything.

The evidence supports institutional continuity rather than full democratization. Part I supports a Victorian information and access club. Part II supports a British access network, where the evidence shows mediated access more clearly than early information. The full paper supports continuity of structure, not sameness of membership. It also leaves profit unproven. Early access is documented. Clear insider profit is not. The listed-price test comes back empty, and the Rothschild Archive warns that this is exactly where insider stories are often invented.

19 Conclusion

The evidence gives a careful answer to the paper's two questions. In the Victorian period, the City of London did work as a closed information and access club. A narrow and recurring set of intermediaries sat near the front of every crisis. The Bank's records showed strain before the public rupture. The telegraph made prices travel faster, but it did not open the circle around the Bank.

The access structure also survived the twentieth century by changing shape. The Bank kept managing stress through recognised intermediaries, from the bill brokers it first gave a facility to in 1830, through the discount and acceptance houses of 1906 and 1931, through the clearing-bank lifeboat of the 1970s, to the seven authorised counterparties of 1996. The continuity index stays high across that whole span. The score for exclusive discount-house dealing falls to zero only at the 1997 reform, when the Bank widened its operations from the recognised discount houses to a larger defined set of eligible counterparties rather than to the general public.

The strongest defensible conclusion is not that the same firms ran British finance for 150 years. They did not. The stronger point is that recognised intermediaries remained the route through which the Bank managed money-market stress until 1997. A few names carried across the wars,

but the durable feature was the actor type, not the firm. What remains uncertain is the firm-level detail of the middle decades, where the surviving evidence is often aggregate, and whether early access ever became profit. The price evidence cannot show that, so the paper does not claim it. The argument appears when separate archives are put on one timeline. Read that way, the British access network was remarkably durable. The club did not simply end. Its structure was carried forward and then written into formal rules.

A Reproducing this paper

- `src/v6_access_network.py` builds every Part II table (`outputs/tables/v6_*.csv`) and Figures 4 to 10, and computes the 1986 to 1996 deposits series.
- `src/v7_flow_network.py` builds the access-network graph (Figure 2b, Table 3b, the centrality and edge tables) with NetworkX and the per-crisis information-flow sequence, and writes the data behind the website’s two interactive panels.
- Part I reuses the project’s Victorian outputs (BoE LOLR ledgers and Bank Rate, the press clocks, the inner-circle access scores, the Yale IMM index); Figures 1 to 3 are produced by `src/v5_figures.py`.

References (read locally unless marked)

- Anson, M., Bholat, D., Kang, M., & Thomas, R. (2017). *The Bank of England as Lender of Last Resort*. Bank of England Working Paper 691.
- Bagehot, W. (1873). *Lombard Street: A Description of the Money Market*.
- Diamond, D. W., & Dybvig, P. H. (1983). “Bank Runs, Deposit Insurance, and Liquidity.” *Journal of Political Economy* 91(3).
- Gorton, G., & Ordoñez, G. (2014). “Collateral Crises.” *American Economic Review* 104(2). (NBER WP 17771.)
- Flandreau, M., & Ugolini, S. (2011). *Where it All Began ...the Overend-Gurney Panic of 1866*. Norges Bank WP 2011/03.
- Hopkins, A. G. (1988). “British Imperialism: A Review and a Revision.” *ReFRESH* 7. (Open-access precursor to Cain & Hopkins, *British Imperialism*, 1993, access-restricted.)
- Amini, S., & Toms, S. (2021). “Elite directors, London finance, and British overseas expansion ...1860 to 1900.” *Economic History Review* 74(2).
- Hoag, C. (2006). “The Atlantic Telegraph Cable and Capital Market Information Flows.” *Journal of Economic History* 66(2). (Author abstract read, body paywalled, headline only.)
- Fessenden, H. (2018). “The Great Telegraph Breakthrough of 1866.” *Econ Focus* (Federal Reserve Bank of Richmond), Q2 2018.
- Schneider, S. (2022). “The politics of last resort lending and the Overend & Gurney crisis of 1866.” *Economic History Review* 75(2).
- Cathcart, B. (2014). “Nathan Rothschild and the Battle of Waterloo.” *The Rothschild Archive: Review of the Year 2013 to 14*.
- Accominotti, O., Lucena-Piquero, D., & Ugolini, S. (2025). “The Origination and Distribution of Money Market Instruments: Sterling Bills of Exchange during the First Globalisation.” *Economic History Review* (in press, read via arXiv:2103.01558).
- Römer, M. (2025). “Financial crisis of 1931? British banking stability and the role of open-market operations.” *Economic History Review* 78(4):1180 to 1201.
- Bank of England (1967). “The London discount market: some historical notes.” *Quarterly Bulletin* 1967.

- Bank of England (1978). “The secondary banking crisis and the Bank of England’s support operations.” *Quarterly Bulletin* 1978 Q2.
- Bank of England statistical database, discount houses 1976 to 1996 (balance-sheet series, data obtained).
- Bank of England, *A Millennium of Macroeconomic Data* (daily Bank Rate).
- Yale International Center for Finance, *Investor’s Monthly Manual* (financial-sector index).
- Not cited at page level: the underlying archival datasets behind Accominotti et al. and Römer.